Quasi-Projective Synchronization of Nonidentical Fractional-Order Neural Networks with Inconsistent Orders

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Published in Physica Scripta 100(1), 015256 (2024) | DOI: 10.1088/1402-4896/ad9c26

Abstract

This paper investigates quasi-projective synchronization for nonidentical fractional-order quaternion-valued neural networks possessing inconsistent fractional derivative orders alpha != beta. Error bound estimations and Lyapunov stability conditions are derived without requiring equal derivative orders across drive and response nodes.

1. Introduction

In traditional synchronization literature, drive and response systems are commonly assumed to share the identical fractional order alpha. However, in heterogeneous networked physical systems, interacting neural components inevitably possess disparate memory lengths and distinct derivative orders.

Because systems with inconsistent orders cannot achieve exact zero-error synchronization (e(t) -> 0), the appropriate qualitative formulation is quasi-synchronization, wherein the synchronization error converges to a small residual bound around zero.

2. Preliminaries and Order Inconsistency Problem

Let the drive system have Caputo order alpha in (0, 1) and the response system have Caputo order beta in (0, 1) with alpha != beta. The state equations are defined over the quaternion field H.

Definition 2. *2 (Quasi-Projective Synchronization with Error Bound). The drive-response network is said to achieve quasi-projective synchronization with error bound epsilon > 0 if there exists a scaling matrix M such that limsup_{t -> infinity} ||y(t) - M x(t)|| <= epsilon.*

Assumption 2. *3 (Bounded Delays and Inconsistent Orders). The transmission delay tau(t) satisfies 0 <= tau(t) <= tau_max and dot{tau}(t) <= d < 1. Furthermore, the difference |alpha - beta| is strictly bounded by delta_0 < 0.2.*

Lemma 2. *1 (Fractional Order Mismatch Decomposition). For functions with orders alpha and beta, the mismatch term can be bounded as ||^C D^beta_t x(t) - ^C D^alpha_t x(t)|| <= C_0 t^{alpha - beta - 1} ||x(t)|| + Delta_m.*

3. Error Bound Estimation and Synchronization Criteria

By constructing a generalized Lyapunov-Krasovskii functional in quaternion domain, we quantify the asymptotic error bound.

Theorem 3. *1 (Ultimate Boundedness of Synchronization Error). Under Assumptions 2.1 and 2.3, if there exists a positive definite matrix P and controller gain K such that the LMI Xi < 0 is satisfied, then the error e(t) = y(t) - M x(t) converges into the compact residual set Omega_epsilon = {e in H^n : ||e|| <= epsilon = sqrt(2 Delta_m / (lambda_min(P) rho))}, where rho is the minimum eigenvalue of the dissipativity matrix.*

Proposition 3. *3 (Optimal Scaling Matrix). The minimal error bound epsilon* is achieved when the scaling matrix M satisfies M = argmin_M ||(beta - alpha) A + [K, M]||_2.*

Remark 3. *5. This framework generalizes previous results which strictly required alpha = beta. Even under order mismatch of up to 15%, the residual error remains confined within a millimeter-scale attractor.*